

Papers whose results conflict with
Douna et al. (2018), *Microquasars as heating sources
of the
intergalactic medium during reionization of the
Universe*

A provenance-tracked literature audit

Prepared for L. Carvalho

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1 Scope, method and audit trail

1.1 Target paper

Douna, V. M., Pellizza, L. J., Laurent, P. & Mirabel, I. F. (2018), *Microquasars as heating sources of the intergalactic medium during reionization of the Universe*, MNRAS **474**, 3488–3499.

DOI: [10.1093/mnras/stx2983](https://doi.org/10.1093/mnras/stx2983) — arXiv:[1711.07374](https://arxiv.org/abs/1711.07374) — bibcode 2018MNRAS.474.3488D.

Local file: Douna (2018) - Microquasars as heating sources... (parent folder).

1.2 How the conflicting set was assembled

Four steps, all executed on 8 September 2026:

1. **Full read of the target paper.** Sections 1–6 of the published MNRAS version were read in full and reduced to the list of testable claims in Section 2.
2. **In-folder search.** All papers in the parent folder were text-searched for statements in conflict with those claims. Findings in Section 4.
3. **Web search — arXiv.** The arXiv API was queried across `astro-ph.HE`, `astro-ph.CO` and `astro-ph.GA` on the axes: cosmic-ray heating/ionization of the IGM; cosmic-ray transport and magnetic confinement at high z ; X-ray-binary contribution to reionization; the metallicity– L_X scaling; 21-cm observational constraints on IGM temperature. Missing bibliographic data were resolved through Crossref.
4. **Web search — NASA ADS.** The ADS API was queried with a user-supplied token for (a) the authoritative citation list `citations(bibcode:2018MNRAS.474.3488D)`, (b) full-text searches (`full:"Douna"`, `full:"microquasar"` `full:"reionization"`), and (c) refereed topical searches on the same axes as step 3.
5. **Screening.** 19 candidates were downloaded; 15 were retained in `against/` and 4 were moved to `screened_no_conflict/` after full-text inspection failed to yield a conflicting statement (Section 5).

1.3 What the ADS query changed (recall audit)

The ADS cross-check materially revised two things.

1. **The citation count was wrong. ADS reports 18 citing papers, not 14.** Semantic Scholar, OpenAlex, OpenCitations and INSPIRE-HEP jointly missed four items, all *Boletín de la Asociación Argentina de Astronomía* proceedings with no DOI and no arXiv posting, which is why no Crossref-derived index carries them:

- Escobar, Pellizza & Romero 2021, BAAA 62, 265, *Microcuásares como fuentes de rayos cósmicos* (2021BAAA...62..265E);
- Garate Núñez, Escobar, Pellizza & Bosch-Ramon 2021, BAAA 62, 234, *Reionización por jets de núcleos galácticos activos* (2021BAAA...62..234G);
- Carvalho, Escobar & Pellizza 2024, BAAA 65, 240, *Cosmic rays at the epoch of reionization* (2024BAAA...65..240C);
- Pellizza, Badaracco, Carvalho, Escobar & Bignone 2025, BAAA 66, 377, *X-ray binary populations in local galaxies* (2025BAAA...66..377P).

All four are now downloaded into the parent folder. [note] The last two are your own group's work with L. J. Pellizza (Douna's co-author); they continue the line rather than conflict with it, and none of the four contradicts any claim in Section 2.

2. **Two new conflicting papers were found**, one of which is now the strongest entry in the catalogue (Dhandha et al. 2025, §3.1) and one moderate (Kaur et al. 2022, §3.11). ADS full-text search also surfaced three papers that turned out to *support* Douna et al.'s premises and were reclassified (Section 5).

[recall, after the cross-check] The four open indices had *complete* recall on refereed, DOI-bearing literature and *zero* recall on Spanish-language national proceedings. Anything comparable in, e.g., RevMexAA or IAU regional proceedings would have been missed before this step.

1.4 Reading conventions used in this document

- Every entry is tagged **HYPOTHESIS** when the conflict is with material in Douna et al.'s Introduction (§1) or Model/Simulations (§2), and **RESULTS** when it is with material in their Results (§3–5) or Discussion and Conclusions (§6).
- **Excerpts are verbatim** from the PDF stored in `against/`. Text was extracted with `pdftotext`; mathematical symbols that the extractor mangled have been restored to their printed form and every such restoration is flagged in the entry. No wording has been altered.
- **Conflict strength** is graded DIRECT (the paper's own stated result is logically incompatible with a Douna claim), MODERATE (incompatible under stated assumptions, or contradicts an input rather than an output), and INDIRECT (undercuts a premise or the motivating context without addressing the claim itself).

[Confidence, stated up front] **None of the 15 papers in `against/` cites Douna et al. (2018), and none is written as a rebuttal of it.** Every conflict below is a conflict I have identified between independently obtained results; it is not a disagreement the authors themselves declare. This is the single most important caveat in this document and it applies to every entry. See Section 6 for the corollary.

2 The claims of Douna et al. (2018) under test

2.1 Hypotheses, premises and modelling choices (§1–§2)

- H1** X-ray binaries, and hence microquasars (MQs), were more numerous and more luminous at high z because of the low metallicity of their parent populations (§1, citing Mirabel et al. 2011; Fragos et al. 2013; Douna et al. 2015; Brorby et al. 2016).
- H2** The X-ray channel is insufficient: “recent works suggest that the contribution of the X-rays emitted by these systems to heating and ionization of the IGM during the EoR is marginal at most (Madau & Fragos 2017)” (§1). This is the stated motivation for turning to CR electrons.
- H3** CR *electrons* are the carrier worth modelling first, “as these particles are expected to make the largest contribution (Tueros et al. 2014)” (§1, final paragraph).
- H4** The reference (“warm”) scenario assumes a pre-heated IGM at $T_{\text{IGM}} \sim 10^4$ K at $z = 10$, which fixes the case-B recombination coefficient $\alpha_{\text{B}} \approx 2.6 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ used throughout §5.1.
- H5** The source is a single MQ (or population) with a power-law electron spectrum, index $\alpha \in [2, 3]$ (fiducial 2.5), total jet kinetic luminosity $L_{\text{k}} = 10^{40} \text{ erg s}^{-1}$, cut-offs free in 10 keV–1 PeV (§5).
- H6** **Magnetic fields are omitted.** “there are still many uncertainties in the knowledge of these fields in the early Universe... Therefore, we will treat magnetic fields separately in a companion paper” (§2.1). Consequently there is no synchrotron cooling and no diffusive confinement; transport is rectilinear.
- H7** EBL, photoionization by secondary photons and galactic photon fields are omitted; the authors argue each omission is *conservative* (i.e. their rates are lower bounds) (§2.1, §6).
- H8** ISM and IGM are cold, homogeneous, partially ionized primordial plasmas; $n_{\text{ISM}} = 1 \text{ cm}^{-3}$, $n_{\text{IGM}} = 2.4 \times 10^{-4} \text{ cm}^{-3}$; cosmological redshifting of the CMB during propagation is neglected (§2.1, §2.3).

2.2 Results and conclusions (§3–§6)

- R1** Low-energy electrons (up to several tens of keV), which carry most of the ionizing power, cannot escape primeval galaxies; any effective mechanism must be two-stage (§6, point i).
- R2** The escape fraction depends on energy *and* on source spectrum and may exceed unity, because cooling creates new electrons (§6, point ii).
- R3** Under the warm scenario, MQs alone can maintain $f_{\text{ion}}^{\text{IGM}} \sim 0.1$ within a few kpc of the galaxy; ionization and heating rates fall as r^{-2} (§5.1, §6 point v).
- R4** Extrapolated to a homogeneous population, the mean ionization rate is $dn_{\text{ion}}/dt \sim 8 \times 10^{-18} \text{ m}^{-3} \text{ s}^{-1}$, “of the order of the typical estimates of Leite et al. (2017) for CRs from SNe” ($2 \times 10^{-18} \text{ m}^{-3} \text{ s}^{-1}$) but far below Pop II UV ($\sim 10^{-14}$) (§6).
- R5** The heating rate is $d\text{Heat}/dt \sim 8 \times 10^{-20} \text{ keV m}^{-3} \text{ s}^{-1}$, corresponding to a global $\Delta T \sim 50$ K at $z \sim 10$ — of the same order as Leite et al. (2017) for CRs and Madau & Fragos (2017) for X-rays (§6).
- R6** Soft spectra are better ionizers and heaters; rates are insensitive to the high-energy cut-off and sensitive to the low-energy one (§5.1, §6 point vi).

R7 Only ~ 5 per cent of the source power is deposited in the IGM inside 1 Mpc; ~ 60 per cent leaves the 1 Mpc volume (§5.3).

3 Catalogue of conflicting papers (folder against/)

Ordered by conflict strength, then by recency.

3.1 Dhandha, Fialkov, Gessey-Jones, et al. (2025) — **HYPOTHESIS** — **DIRECT** (strongest entry)

Dhandha, J., Fialkov, A., Gessey-Jones, T., et al. 2025, *Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, MNRAS **544**, 1608. DOI: [10.1093/mnras/staf1736](https://doi.org/10.1093/mnras/staf1736); arXiv:2508.13761; bibcode 2025MNRAS.544.1608D. Found via ADS full-text search; absent from the arXiv-only sweep.

File: against/Dhandha (2025) - Narrowing the discovery space... [arXiv 2508.13761].pdf

Conflicts with: **H4** (the warm, $T_{\text{IGM}} \sim 10^4$ K IGM at $z = 10$), and through it **R3**. This is the same conflict as the HERA entry below, but stated at Douna et al.’s exact fiducial redshift and with a tighter bound.

Verbatim excerpt (Abstract)

“From CXB and HERA data, we infer the IGM kinetic temperature to be $T_{\text{K}}(z = 15) \lesssim 7.7$ K, $2.5 \text{ K} \lesssim T_{\text{K}}(z = 10) \lesssim 66$ K, and $20 \text{ K} \lesssim T_{\text{K}}(z = 6) \lesssim 2078$ K at 95% credible interval (CI).”

Analysis. Douna et al. set $z = 10$ as their working redshift (§2.1) and evaluate the recombination rate their electrons must outcompete at $T_{\text{IGM}} \sim 10^4$ K (§5.1). Dhandha et al. infer $T_{\text{K}}(z = 10) \lesssim 66$ K at 95% credibility by combining the cosmic X-ray background, HERA power-spectrum upper limits, the SARAS 3 non-detection and HST/JWST UV luminosity functions. That is a factor ~ 150 below the assumed value, and it brackets $z = 10$ directly rather than by interpolation. It sits much closer to Douna et al.’s own “extremely cold” comparison case ($T \sim 30$ K, the CMB temperature at $z = 10$), for which they state the total recombination coefficient is “around 100 times higher than the case B recombination coefficient”. On the observationally favoured branch, therefore, the recombination rate their electrons compete against is roughly two orders of magnitude larger than in the scenario from which **R3** ($f_{\text{ion}} \sim 0.1$ within a few kpc) is quoted.

Contradictory data / ambiguity. [explicit] Four qualifications, all material. (i) **This is an inference, not a measurement.** It is a posterior from a Pop II galaxy model conditioned on several data sets, and the authors close their own abstract with the caveat: “the results depend on our assumption of a redshift-independent X-ray and radio efficiency of galaxies and on the exclusion of a flexible model for Population III stars.” Since Douna et al.’s sources are *explicitly* Pop III-era microquasars, the excluded model class is the one her scenario inhabits. This weakens the entry considerably and must be quoted alongside the bound. (ii) It leans on the SARAS 3 non-detection (§3.13), itself contested. (iii) It cites Douna et al. (2015) — the metallicity paper — *in support* of the strong metallicity dependence, i.e. it corroborates **H1** while contradicting **H4**. (iv) T_{K} is a volume-averaged quantity. Douna et al.’s **R3** is a statement about gas within a few kpc of a source, which a global mean does not directly constrain. **I did not recompute the local-versus-global correction and this remains the weakest link in the argument.**

3.2 HERA Collaboration (2023) — **RESULTS** and **HYPOTHESIS** — DIRECT

Abdurashidova, Z., Adams, T., Aguirre, J. E., et al. (HERA Collaboration), 2023, *Improved Constraints on the 21 cm EoR Power Spectrum and the X-Ray Heating of the IGM with HERA Phase I Observations*, ApJ **945**, 124. DOI: [10.3847/1538-4357/acaf50](https://doi.org/10.3847/1538-4357/acaf50); arXiv:2210.04912.

File: against/HERA Collaboration (2023) - Improved Constraints... [arXiv 2210.04912].pdf

Conflicts with: **H4** (the warm, $T_{\text{IGM}} \sim 10^4$ K IGM at $z = 10$) and, through it, **R3** (which depends on α_B evaluated at that temperature).

Verbatim excerpt (§7, “Constraints on the IGM”)

“The new HPD 95% credible intervals on the spin and kinetic temperatures are $4.7 \text{ K} < T_S < 171.2 \text{ K}$ and $3.2 \text{ K} < T_K < 313.2 \text{ K}$ at $z = 10.4$ and $15.6 \text{ K} < T_S < 656.7 \text{ K}$ and $13.0 \text{ K} < T_K < 4768 \text{ K}$ at $z = 7.9$.”

Second excerpt (Abstract)

“we find that the intergalactic medium must have been heated above the adiabatic cooling limit at least as early as $z = 10.4$, ruling out a broad set of so-called ‘cold reionization’ scenarios. If this heating is due to high-mass X-ray binaries during the cosmic dawn, as is generally believed, our result’s 99% credible interval excludes the local relationship between soft X-ray luminosity and star formation”

Analysis. Douna et al. evaluate the recombination rate that their electrons must outcompete at $T_{\text{IGM}} \sim 10^4$ K (§5.1). HERA’s 95% upper bound at $z = 10.4$ — the redshift closest to Douna’s fiducial $z = 10$, and slightly *later*, hence if anything warmer — is $T_K < 313.2$ K, i.e. lower than the assumed value by a factor $\gtrsim 30$. Douna et al. themselves state the consequence: “A colder medium would imply a higher recombination rate, and thus a lower reionization power of electrons” (§5.1). Their own “cold” bracket uses $T \sim 30$ K, where the recombination coefficient is “around 100 times higher than the case B recombination coefficient”. The observationally permitted temperature at $z \approx 10$ therefore sits in the lower part of their bracketed range, not at the warm end on which R3 ($f_{\text{ion}} \sim 0.1$ within a few kpc) is quoted. This entry is retained alongside Dhandha et al. (§3.1) because it is the primary observational result: Dhandha et al. tighten the bound and place it at $z = 10$ exactly, but only through an inference layer that HERA’s own limit does not require.

Contradictory data / ambiguity. [explicit] This same paper cuts the other way on **H2**. HERA *requires* heating above adiabatic and excludes the local L_X –SFR relation, which is an argument that early accreting binaries were *more*, not less, effective — supporting Douna’s H1 while undermining her H2 (that X-rays are “marginal at most”). The paper also states of the weak-X-ray exclusion that “that statement is somewhat model dependent”. So: HERA contradicts H4 directly and H2 directly, but in doing so it strengthens H1. This entry must not be cited as a blanket refutation.

3.3 Leite, Evoli, D’Angelo, Ciardi, Sigl & Ferrara (2017) — **RESULTS** — DIRECT

MNRAS **469**, 416–424. DOI: [10.1093/mnras/stx805](https://doi.org/10.1093/mnras/stx805); arXiv:1703.09337.

File: against/Leite (2017) - Do Cosmic Rays Heat the Early Intergalactic Medium [arXiv 1703.09337].pdf

Conflicts with: **R3** and **R4** (the ionization claims).

Verbatim excerpt (Abstract)

“we show that CRs contribute negligibly to IGM ionization, but heat it substantially, raising its temperature by $\Delta T = 10 - 200$ K by $z = 10$, depending on the CR injection spectrum.”

Second excerpt (§4.1, “Impact of CRs on reionization”)

“The CR ionization rate is several orders of magnitude smaller than the UV photoionization rate. This justifies the fact that CR were not included when we described our reionization model in §2.2.”

Analysis. This is the single sharpest conflict in the audit, and it is internal to Douna et al.’s own benchmarking. In §6 they write that their extrapolated ionization rate is “of the order of the typical estimates of Leite et al. (2017) for CRs from SNe” — and then conclude (§6, point v) that MQs “would be able to maintain by themselves large ionization fractions (of the order of 0.1) near the galaxies”. Leite et al., for a rate of the same order, conclude the opposite: that the CR contribution to ionization is negligible and can be dropped from the reionization model entirely. The two papers agree on the number and disagree on what it means.

Contradictory data / ambiguity. [explicit] Three qualifications. (i) **Chronology:** Leite et al. (2017) precedes Douna et al. (2018) and is cited by them; this is a tension Douna et al. were aware of and did not resolve, not a subsequent refutation. (ii) **Carrier:** Leite et al. model CR *protons* from Pop II supernovae; Douna et al. model CR *electrons* from MQ jets. The rates are compared by the original authors, so the comparison is theirs, but the physical systems differ. (iii) **Scale:** Leite et al. compare against the *global* UV photoionization rate, whereas Douna et al.’s $f_{\text{ion}} \sim 0.1$ claim is explicitly local (within a few kpc of the source). The two statements are not strictly mutually exclusive; the conflict is in the interpretation of significance, not in arithmetic. I could not resolve this cleanly and flag it as unresolved.

3.4 Gessey-Jones, Fialkov, de Lera Acedo, Handley & Barkana (2023) — **RE-SULTS** — DIRECT

MNRAS **526**, 4262–4284. DOI: [10.1093/mnras/stad3014](https://doi.org/10.1093/mnras/stad3014); arXiv:2304.07201.

File: against/Gessey-Jones (2023) - Signatures of Cosmic Ray Heating in 21-cm Observables [arXiv 2304.07201].pdf

Conflicts with: R4, R5 and R7 (the extrapolation from kpc-scale deposition to a homogeneous, cosmologically relevant mean rate).

Verbatim excerpt (§5, Discussion and Conclusions)

“We find the spatial range of cosmic ray heating to be much shorter than that of X-ray heating, even when cosmic rays are assumed to travel along straight lines. This short-range nature manifests in the IGM heating being clustered around regions of efficient star formation and results in a sharp contrast in the 21-cm signal between localized heated regions of emission and vast unheated regions of absorption.”

Analysis. The clause “even when cosmic rays are assumed to travel along straight lines” is decisive, because rectilinear propagation is precisely Douna et al.’s transport model (H6). Douna et al. nonetheless convert a kpc-scale, r^{-2} deposition profile into a single homogeneous mean rate (R4, R5) by imagining “a set of homogeneously distributed galaxies containing MQs”. Gessey-Jones et al. show that this averaging step is exactly what CR heating does not permit: the medium separates into hot islands and vast cold voids, so a volume-averaged $\Delta T \sim 50$ K does not describe the thermal state of the IGM anywhere.

Contradictory data / ambiguity. [explicit] Two, both material. (i) The same paper reports that “cosmic rays with much higher energies up to 200 MeV have a non-negligible contribution, heating up the IGM on large scales out to ~ 100 cMpc away from their sources” — so the short-range conclusion is spectrum-dependent, and Douna et al.’s hard tail (up to 1 PeV) is not covered by it. (ii) Gessey-Jones et al. model **CR protons only**; they explicitly set CR electrons aside (footnote 1: “Cosmic ray electrons are however important to consider when modelling SNR-produced radio backgrounds”). Their transport conclusion is therefore not a computation about Douna et al.’s carrier. Grade DIRECT is assigned because the geometric argument is carrier-blind, but this is a judgement call, not a demonstration.

3.5 Owen, Jacobsen, Wu & Surajbali (2018) — HYPOTHESIS — DIRECT

MNRAS **481**, 666–687. DOI: [10.1093/mnras/sty2279](https://doi.org/10.1093/mnras/sty2279); arXiv:1808.07837.

File: against/Owen (2018) - Interactions between Ultra-High-Energy Particles and Protogalactic Environments [arXiv 1808.07837].pdf

Conflicts with: **H6** (omission of magnetic fields) and, as a consequence, **R1**, **R2** and **R7** (escape fractions and the energy budget leaving the galaxy).

Verbatim excerpt (Abstract)

“We have found that protogalactic magnetic fields and their evolution can significantly affect the energy transport and energy deposition processes of cosmic rays. Within a Myr after the onset of star-formation the magnetic field in a protogalaxy could attain a strength sufficient to confine all but the highest energy particles within the galaxy.”

Analysis. Douna et al. defer magnetic fields to a companion paper and compute rectilinear escape from a 1 kpc galaxy. If a protogalaxy magnetizes within ~ 1 Myr to the point of confining all but the highest-energy particles, then the escape fractions of R1–R2 — and with them the ~ 65 per cent of injected power that Douna et al. report as reaching the IGM (R7) — are computed in a regime that does not obtain. Note the direction: confinement would suppress the escape of exactly the intermediate-energy electrons that Douna et al. identify as the reservoir feeding the low-energy ionizing population (R2, R6).

Contradictory data / ambiguity. [explicit] The same abstract preserves a window for Douna et al.’s scenario: “there is a window before the protogalaxy is fully magnetised, energetic particles could stream across the galaxy freely, delivering energy into the circumgalactic and intergalactic medium.” Whether MQ jets operate inside or outside that window is not determined by either paper. Owen et al. also model *hadronic* particles; the confinement argument is magnetic-rigidity-based and so applies to electrons of the same rigidity, but the paper performs no electron calculation.

3.6 Kaur, Qin & Mesinger (2022) — HYPOTHESIS — MODERATE

MNRAS **513**, 5097. DOI: [10.1093/mnras/stac1226](https://doi.org/10.1093/mnras/stac1226); arXiv:2203.10851; bibcode 2022MNRAS.513.5097K.

Found via ADS topical search.

File: against/Kaur (2022) - The 21-cm signal from the cosmic dawn - metallicity dependence... [arXiv 2203.10851].pdf

Conflicts with: **H1** — not its direction, but the precision with which Douna et al. deploy it.

Verbatim excerpt (Abstract)

“For our fiducial models, galaxies with star formation rates of order 10^{-3} – $10^{-1} M_{\odot} \text{ yr}^{-1}$ and metallicities of order 10^{-3} – $10^{-2} Z_{\odot}$ are the dominant contributors to the X-ray background (XRB) during this period. Different $L_X/\text{SFR}-Z$ relations result in factors of ~ 3 differences in these ranges, as well as in the mean IGM temperature and the large-scale 21-cm power, at a given redshift.”

Analysis. Douna et al. invoke the metallicity enhancement (H1) qualitatively and then adopt a single fiducial $L_k = 10^{40} \text{ erg s}^{-1}$ with no uncertainty band. Kaur et al. show that merely *choosing between published $L_X/\text{SFR}-Z$ relations* moves the mean IGM temperature by a factor ~ 3 . Since all of Douna et al.’s rates scale linearly with L_k (§5), an unquantified factor of a few propagates directly into R3–R5. Together with Sartorio et al. (§3.10), which adds an IMF-driven spread of orders of magnitude, this is the case for attaching an error bar to the fiducial luminosity rather than quoting single-valued rates.

Contradictory data / ambiguity. [explicit] Kaur et al. model the *X-ray* output of HMXBs, not jet kinetic power; the factor-3 spread is not automatically the spread in L_k . The paper affirms the metallicity dependence itself, so this contradicts the treatment of H1 as a precise input, not H1 as a claim.

3.7 Jana & Nath (2018) — RESULTS — MODERATE

MNRAS **479**, 153–161. DOI: [10.1093/mnras/sty1481](https://doi.org/10.1093/mnras/sty1481); arXiv:1806.01295.

File: against/Jana (2018) - Cosmic ray heating of intergalactic medium - patchy or uniform [arXiv 1806.01295].pdf

Conflicts with: R3 (the localized, kpc-scale picture) and H6/H8 (rectilinear transport through a neutral IGM).

Verbatim excerpt (§5, Summary)

“In the case of CRs from mini-haloes at $z \sim 10$ –20, we put an upper bound on the diffusion coefficient ($D \leq 1 \times 10^{26} \text{ cm}^2 \text{ s}^{-1}$ for $z \sim 10$ and $D \leq 5\text{--}6 \times 10^{26} \text{ cm}^2 \text{ s}^{-1}$ for $z \sim 20$) for which the heating is inhomogeneous, after comparing our temperature profiles with the estimate of global temperature increase. Given the expected scaling of the diffusion coefficient with redshift, this bound suggests uniform heating, both at $z \sim 10$ and 20.”

[transcription] The extractor rendered the printed “ $\leq$ ” as the digit “6”; restored above. Exponents were reconstructed from the printed form (“ $1 \times 10^{26} \text{ cm}^2 \text{ s}^{-1}$ ”).

Second excerpt (§5, Summary)

“The surrounding medium of galaxies at high redshift is likely to be ionized on account of star formation in the galaxy. Therefore, the heating by CRs will proceed in a different manner than previously considered case of neutral IGM heating.”

Analysis. Douna et al.’s central spatial claim (R3) is that the interesting action is *local* — $f_{\text{ion}} \sim 0.1$ within a few kpc, falling as r^{-2} . Jana & Nath, using the diffusion coefficient expected from the scaling $D \propto B^{-1/3}$ with a $\sim 10^{-9} \text{ G}$ IGM field, conclude that CR heating at $z \sim 10$ should be *uniform*, not patchy. Their second excerpt also conflicts with the framing of H8: they argue the near-galaxy medium is already photoionized, so the neutral-IGM energy-partition calculation Douna et al. perform is not the applicable one.

Contradictory data / ambiguity. [explicit] The authors immediately disclaim their own conclusion: “But the uncertainties in CR parameters (spectrum, lower energy limit of emerging CRs, diffusion coefficient and its dependence on magnetic field) and magnetic field at high redshift precludes any firm conclusion.” Note also that Jana & Nath and Gessey-Jones et al. (previous entry) reach *opposite* conclusions on patchy-vs-uniform. This is an open disagreement in the literature, not a settled result against Douna et al.; what both share is that neither supports her combination of local structure *and* a meaningful global average. Jana & Nath’s first summary bullet — “It is not easy for low energy CRs to escape relatively massive galaxy” — *agrees* with Douna et al.’s R1.

3.8 Yokoyama & Ohira (2023) — **HYPOTHESIS** (model) — MODERATE

MNRAS **523**, 3671–3677. DOI: [10.1093/mnras/stad1596](https://doi.org/10.1093/mnras/stad1596); arXiv:2302.07028.

File: against/Yokoyama (2023) - Resistive Heating Induced by Streaming Cosmic Rays Around a Galaxy in the Early Universe [arXiv 2302.07028].pdf

Conflicts with: H6 and the process inventory of §2.1, and hence with the mechanism attribution behind R5.

Verbatim excerpt (§4, Summary)

“We revisited the influence of streaming CRs on the heating of the IGM. We clarified that resistive heating induced by CRs becomes dominant over the other mechanisms in the vicinity of a galaxy at $r \lesssim 10^2$ kpc in early phases. The temperature reaches 10^4 K at $r = 1$ kpc in 2×10^6 yr.”

Analysis. Douna et al. enumerate their heating channels explicitly in §2.1: collisional ionization and excitation, elastic e^-e^- scattering, bremsstrahlung, inverse Compton and direct Compton. Yokoyama & Ohira identify a channel absent from that list — Ohmic dissipation of the thermal return current driven by the streaming CR beam — and find it dominant over the collisional channels precisely in the spatial range ($r \lesssim 100$ kpc) where Douna et al. compute their profiles. The attribution of the heating in R5 to e^-e^- scattering is therefore incomplete for a streaming population.

Contradictory data / ambiguity. [explicit — direction matters] This conflict does *not* weaken Douna et al.’s conclusion; it points the same way with a different mechanism, and would if anything raise the heating rate. It is included because the user’s criterion covers conflicts with the model as well as with the numbers, and because it falsifies Douna et al.’s claim (H7, §6) that their omissions are uniformly “conservative” in the sense of being physically complete-but-underestimated: a missing dominant channel is a different kind of gap. Note also that resistive heating requires a streaming beam and a return current, i.e. a magnetized medium — the very ingredient Douna et al. set aside. Yokoyama & Ohira model **CR protons**.

3.9 Owen, Jin, Wu & Ferreras (2019) — **RESULTS** — MODERATE

MNRAS **484**, 1645–1671. DOI: [10.1093/mnras/stz060](https://doi.org/10.1093/mnras/stz060); arXiv:1901.01411.

File: against/Owen (2019) - Hadronic Interactions of Energetic Charged Particles in Protogalactic Outflow Environments [arXiv 1901.01411].pdf

Conflicts with: R4, R5, R7 (the spatial extent of energy deposition).

Verbatim excerpt (Abstract)

“In a purely diffusive transport scenario, we find the peak heating rate reaches $10^{-26} \text{ erg cm}^{-3} \text{ s}^{-1}$ at the base of the outflow where the wind is driven by core-collapse supernovae at an event rate of 0.1 yr^{-1} , but does not extend beyond 2 kpc. In the advection limit, the peak heating rate is reduced to $10^{-28} \text{ erg cm}^{-3} \text{ s}^{-1}$, but its extent can reach to tens of kpc.”

Analysis. Douna et al. follow electrons to 1 Mpc and report that ~ 60 per cent of the emitted power still leaves that volume (R7). Owen et al. find that, once realistic diffusive or advective transport through a protogalactic outflow is imposed, CR energy deposition terminates at 2 kpc (diffusive) or tens of kpc (advective) — one and a half to three orders of magnitude short of the volume over which Douna et al. compute and then extrapolate.

Contradictory data / ambiguity. [explicit] The comparison is not like-for-like in two respects: Owen et al. treat *hadronic* CRs in a supernova-driven *outflow*, whereas Douna et al. treat electrons injected by a jet into a static homogeneous medium; and the two papers use different energy-loss physics (pion production vs. Coulomb and inverse Compton). The conflict is over what transport regime is appropriate, and neither paper adjudicates it. **This entry was not independently recomputed.**

3.10 Sartorio, Fialkov, Hartwig, et al. (2023) — HYPOTHESIS — MODERATE

MNRAS **521**, 4039–4055. DOI: [10.1093/mnras/stad697](https://doi.org/10.1093/mnras/stad697); arXiv:2303.03435.

File: against/Sartorio (2023) - Population III X-ray Binaries and their Impact on the Early Universe [arXiv 2303.03435].pdf

Conflicts with: **H1** and **H5** (the assumption that a single fiducial $L_k = 10^{40} \text{ erg s}^{-1}$ characterizes the high- z MQ population).

Verbatim excerpt (Abstract)

“Specifically, we find that the uniformity and strength of the X-ray feedback in the intergalactic medium is strongly dependant on the IMF. Bottom-heavy IMFs result in a smoother distribution of XRBs, but have a luminosity orders of magnitude lower than more top-heavy IMFs. Top-heavy IMFs lead to more spatially uneven, albeit strong, X-ray emission.”

Second excerpt (§5.5, Halo Absorption)

“soft photons with energies below 0.1 keV are completely absorbed and, thus, contribute nothing to IGM heating and ionization.”

Analysis. Douna et al. fix $L_k = 10^{40} \text{ erg s}^{-1}$ and note that “all the resulting spectra are linear in this parameter” (§5). Sartorio et al. show that the population-integrated luminosity of primordial XRBs spans *orders of magnitude* depending on the Pop III IMF, which is unconstrained. Because Douna et al.’s results scale linearly with L_k , the uncertainty in H5 propagates linearly into R3–R5, an uncertainty band Douna et al. do not carry. This is a conflict with the premise’s precision, not with its sign.

Contradictory data / ambiguity. [explicit] Sartorio et al. treat the *X-ray* output of Pop III XRBs, not jet kinetic power; the mapping from one to the other is model-dependent and is not made in either paper. Their second excerpt is a conflict with **H2** in the opposite direction from HERA’s: it supports Douna et al.’s premise that the soft X-ray channel is ineffective. **The two excerpts from this single paper therefore cut in opposite directions** and the entry should be cited with that stated.

3.11 Lehmer, Eufrasio, Basu-Zych, et al. (2021) — HYPOTHESIS — MODERATE

ApJ **907**, 17. DOI: [10.3847/1538-4357/abceec1](https://doi.org/10.3847/1538-4357/abceec1); arXiv:2011.09476.

File: against/Lehmer (2021) - The Metallicity Dependence of the HMXB Luminosity Function [arXiv 2011.09476].pdf

Conflicts with: H1 (the metallicity premise, as stated without luminosity qualification).

Verbatim excerpt (Abstract)

“We demonstrate that the SFR-normalized HMXB XLF shows clear trends with metallicity, with steadily increasing numbers of luminous and ultraluminous X-ray sources ($\log L \text{ (erg s}^{-1}\text{)} = 38\text{--}40.5$) with declining metallicity. However, we find that the low-luminosity ($\log L \text{ (erg s}^{-1}\text{)} = 36\text{--}38$) HMXB XLF appears to show a nearly constant SFR scaling and slope with metallicity.”

Analysis. Douna et al. state H1 without restriction: “there is strong evidence that supports that XRBs are more numerous in low-metallicity environments” (§1), citing seven references including Douna et al. (2015). Lehmer et al. show the enhancement is confined to $L_X \gtrsim 10^{38} \text{ erg s}^{-1}$ and is absent below it. The premise survives for the ULX-class systems relevant to $L_k = 10^{40} \text{ erg s}^{-1}$, but not as the general population statement Douna et al. make.

Contradictory data / ambiguity. [explicit] This is a refinement that mostly *supports* Douna et al.’s specific use case, since her fiducial source is at the luminous end. Graded MODERATE, not DIRECT, for that reason. Note that Lehmer et al. (2021 and 2024) cite Douna et al. (2015) — the metallicity paper — as supporting evidence, not the 2018 paper under audit.

3.12 Sotomayor Checa, Romero & Bosch-Ramon (2021) — HYPOTHESIS — MODERATE

Ap&SS **366**, 1. DOI: [10.1007/s10509-020-03911-5](https://doi.org/10.1007/s10509-020-03911-5); arXiv:2012.11664 (posted 2020).

File: against/Sotomayor Checa (2020) - Equatorial outflows driven by jets in Population III microquasars [arXiv 2012.11664].pdf

Conflicts with: H5 / H3 (that a MQ jet delivers a clean power-law electron population into the ISM).

Verbatim excerpt (§5, conclusions)

“The geometry we have adopted for the wind of the accretion disk is a purely axisymmetrical flow. In this way, we can analyze more adequately the production of a jet-induced equatorial wind. However, a more realistic geometry should take into account a wind that interacts collisionally with the jet. This would lead to shocks in the jet and to a decrease in jet velocity, likely reducing the capacity of the jet to emerge from the binary system.”

Second excerpt (§5, conclusions)

“When $L_{\text{jet}} \ll L_{\text{wind}}$ (Case 3) or $L_{\text{jet}} \sim L_{\text{wind}}$ (Case 2), jets keep narrow while the winds do not get significantly deflected sideways. In such a case, the system is a powerful source of UV radiation and soft X-rays emitted by the wind”

Analysis. Relativistic hydrodynamic simulations by the group that built the Pop III MQ model show that in two of three power regimes the system’s output is dominated by wind UV and soft X-rays rather than by jet particles, and that a more realistic jet–wind geometry would likely impede the jet from leaving the binary at all. Douna et al. take the injection of jet electrons into the ISM as given (§2.1, “We assume that electrons are produced in the jets of MQs and injected into the ISM, as discussed by Heinz & Sunyaev 2002”).

Contradictory data / ambiguity. [explicit — weak entry] The conflicting statement is the authors’ *own caveat on their own model*, phrased with “likely”, not a demonstrated result; their abstract concludes the opposite overall (“Population III microquasars might inject gamma rays and relativistic particles into the early intergalactic medium, contributing to its reionization”). This entry is the weakest in the catalogue and should be cited as a modelling caution, not as a contradiction.

3.13 Yang, Long & Hirata (2024) — RESULTS (relevance) — INDIRECT

ApJ **965**, 111. DOI: [10.3847/1538-4357/ad3237](https://doi.org/10.3847/1538-4357/ad3237); arXiv:2311.18721 (posted 2023).

File: against/Yang (2023) - Compton scattering of electrons in the intergalactic medium [arXiv 2311.18721].pdf

Conflicts with: H3 and the broader relevance claim behind R5 (that CR electrons are an energetically significant IGM component).

Verbatim excerpt (Abstract)

“At mean density, the ratio of cosmic ray electron to thermal pressure in the IGM $P_{\text{CRE}}/P_{\text{th}}$ is 0.3% at $z = 2$, rising to 1.0% at $z = 1$, and 1.8% at $z = 0.1$ (considering only the cosmic rays produced locally by Compton scattering).”

Analysis. The only paper found that computes the CR-electron energy budget of the IGM as such. It finds the population sub-percent in pressure relative to the thermal IGM and *decreasing* toward high redshift (energy density peaks at $z \approx 2$).

Contradictory data / ambiguity. [explicit — do not overclaim] This is a weak entry and the mismatch is severe: the calculation runs $z \leq 6$, not $z = 10$; the CR electrons are produced by Compton scattering of the extragalactic background, not injected by jets; and the parenthetical “considering only the cosmic rays produced locally by Compton scattering” explicitly excludes source-injected populations of the kind Douna et al. model. It is included because it is the closest existing quantitative statement on the relevance of IGM CR electrons, and because its redshift trend runs against the assumption that this component grows in importance at cosmic dawn. **No extrapolation of their numbers to $z = 10$ was attempted and none should be inferred.**

3.14 Bera, Samui & Datta (2023) — HYPOTHESIS (attribution) — INDIRECT

MNRAS **519**, 4869–4883. DOI: [10.1093/mnras/stac3814](https://doi.org/10.1093/mnras/stac3814); arXiv:2202.12308 (posted 2022).

File: against/Bera (2022) - Impact of cosmic rays on the global 21-cm signal during cosmic dawn [arXiv 2202.12308].pdf

Conflicts with: H2/H3 (the attribution of the required CR heating to MQ electrons rather than SN protons) and the process inventory of §2.1.

Verbatim excerpt (Abstract)

“The low energy cosmic ray protons from Pop III supernovae can escape from minihalos and heat the IGM via collision and ionization of hydrogen. Furthermore, high energy protons generated in Pop II supernovae can escape the hosting halos and heat the IGM via magnetosonic Alfvén waves. We show that the heating due to these cosmic ray particles can significantly impact the IGM temperature and hence the global 21-cm signal at $z \sim 14 - 18$.”

Analysis. Two points of friction. First, the whole cosmic-dawn CR heating budget is assigned to supernova protons, with no MQ contribution — an attribution conflict with Douna et al.’s framing that MQ electrons are the carrier “expected to make the largest contribution” (H3). Second, the Alfvén-wave (magnetosonic) transfer channel is absent from Douna et al.’s process list (§2.1) and, like the resistive channel of Yokoyama & Ohira, requires the magnetic field she omits (H6).

Contradictory data / ambiguity. [explicit] Bera et al. nowhere assert that MQs are unimportant; they simply do not model them. Absence of a term in someone else’s budget is weak evidence against it. The entry is retained for the missing-channel point, which is concrete.

3.15 Singh, Nambissan T., Subrahmanyan, et al. (SARAS 3; 2022) — **HYPOTHESIS** (context) — INDIRECT

Nature Astronomy **6**, 607–617. DOI: [10.1038/s41550-022-01610-5](https://doi.org/10.1038/s41550-022-01610-5); arXiv:2112.06778 (posted 2021).

File: against/Singh (2022) - SARAS 3 - On the detection of a cosmic dawn signal in the radio background [arXiv 2112.06778].pdf

Conflicts with: the observational context invoked for **H2/H4** — specifically the EDGES-driven case that cosmic dawn requires non-standard heating or radio backgrounds.

Verbatim excerpt (Abstract)

“We report a radiometer measurement of the spectrum of the radio sky in the 55–85 MHz band, which shows that the profile found by Bowman et al. in data taken with the Experiment to Detect the Global Epoch of Reionization Signature (EDGES) low-band instrument is not of astrophysical origin; their best-fitting profile is rejected with 95.3% confidence.”

Analysis. Douna et al. (2018) predate EDGES and do not invoke it, so this is *not* a conflict with their paper as written. It is included because the downstream literature that carries her line of work forward — Mirabel (2019), Mirabel & Rodríguez (2022), both in the parent folder — rests substantially on the EDGES absorption being real. If the user’s paper builds on that thread, this is the constraint that has to be addressed.

Contradictory data / ambiguity. [explicit] The EDGES/SARAS 3 disagreement is unresolved in the literature. A rejection at 95.3% confidence is not a refutation, and the two experiments have not been reconciled. **This entry is context, not evidence against Douna et al.**

4 Conflicts found inside the parent folder

These four papers are already in the parent folder (they are among the 14 that cite Douna et al. 2018) and were *not* duplicated into **against/**. Each contains a statement in conflict with a Douna claim while citing her neutrally or approvingly elsewhere.

Paper	Target	Verbatim conflicting excerpt
Sazonov & Khabibullin (2018), MNRAS 476 , 2530; arXiv:1712.04831	H2	HYPOTHESIS implies that ULXs (and hence HMXBs as a whole) can induce a stronger heating of the IGM in the early Universe compared to estimates neglecting X-ray feedback.” (Summary)
<i>Analysis:</i> Douna et al. motivate the CR-electron channel by asserting the X-ray channel is “marginal at most”. Sazonov & Khabibullin show the soft X-ray escape fraction from a ULX-hosting minihalo rises from ~20–50% to ~30–80% over the ULX lifetime, i.e. X-rays are <i>more</i> effective than the estimates Douna et al. rely on. [ambiguity] The same Summary limits this to $M \lesssim 3 \times 10^8 M_\odot$ haloes and concludes the population-averaged escape fraction “probably remained low, $\lesssim 30\%$, at these epochs”.		
Bosch-Ramon (2018), A&A 617 , L3; arXiv:1808.08911	H3	HYPOTHESIS the most energetic cosmic rays can diffuse out of the radio lobes; otherwise, they stay in the lobes and cool through adiabatic losses. Secondly, it is not known which fraction of the jet energy is in the form of very energetic cosmic rays.” (§1)
<i>Analysis:</i> Written of AGN lobes, but it is the reason Bosch-Ramon abandons escaping CRs and works with <i>in situ</i> lobe electrons radiating via inverse Compton instead. Douna et al. assume escape. [ambiguity] The scale argument does not transfer cleanly from Mpc-scale AGN lobes to pc-scale MQ jets; Bosch-Ramon cites Douna et al. approvingly two sentences earlier.		
Escobar, Pellizza & Romero (2022), A&A 665 , A61; arXiv:2207.08633	H3	HYPOTHESIS focus on the secondary protons product of neutron beta decay. Electrons and neutrinos are also emitted in the process but they carry a negligible fraction of the progenitor neutron energy ($\sim 0.1\%$).” (§2)
<i>Analysis:</i> The same group’s 2021 paper (A&A 650, A136) explicitly set out to “support the claims of Douna et al. (2018)”; by 2022 the electron component has been dropped as energetically negligible in the neutron-escape channel, and the mature version of the mechanism is proton-dominated. [ambiguity] This concerns the neutron-decay channel specifically, not shock-accelerated jet electrons, which is what Douna et al. model.		
Mirabel & Rodríguez (2022), New Astron. Rev. 94 , 101642; arXiv:2203.12741	H1	HYPOTHESIS unreasonable large number of sources of this type that are required to match the onset of the EDGES trough would exceed the expected maximum total mass of stars already formed at that epoch by factors of $\sim 10^6$ for sources like Cygnus X-1” (§X, conclusion 4)
<i>Analysis:</i> A co-author of Douna et al. (2018) arguing that BH-HMXB microquasars of the known Galactic type cannot, in the required numbers, drive the cosmic-dawn signal. [ambiguity] The argument is specifically about matching the EDGES trough onset via a radio background — a different observable from IGM heating — and the paper’s alternative is IMBHs, not the abandonment of the microquasar picture.		

5 Screened, no conflict found

Downloaded, read, and moved to `screened_no_conflict/` because full-text inspection produced no statement in conflict with any claim in Section 2. Listed so the search is auditable.

- **Sazonov & Khabibullin (2017)**, Astron. Lett. **43**, 211; arXiv:1612.01262 — *Preheating of the early universe by radiation from high-mass X-ray binaries*. Found by ADS full-text search and **initially catalogued as a conflict with H2 on the strength of its abstract**, which reads that HMXB X-rays “could significantly heat ... the intergalactic medium by $z \sim 10$ ” under stated conditions. Reading §4 reverses that: “Our conclusion that the Universe could not be significantly heated by X-rays from HMXBs before $z \sim 10$ is similar to that reached by Madau & Fragos (2016).” The paper therefore **supports** Douna et al.’s premise H2 rather than contradicting it, and independently corroborates the Madau & Fragos (2017) result she cites. [This is a worked example of why abstract-level screening is not sufficient; the conditional in the abstract

inverts in the conclusions.]

- **Lehmer, Monson, Eufrasio, et al. (2024)**, arXiv:2410.19901 — updated metallicity/SFH framework for XRB populations. Cites Douna et al. (2015) as *supporting* the L_X /SFR–metallicity anti-correlation. No conflict; supersedes Lehmer et al. (2021) without reversing it.
- **Leong & Meiksin (2025)**, arXiv:2509.10255 — 21-cm signature of X-ray heated haloes. Its “most of the heating will be uniform on measurable scales” reads as a conflict with Douna et al.’s localized picture, but the paper’s substance (bright sources produce detectable warm rings) *supports* the existence of a local heating zone. Net: no clear conflict. [judgement call — reconsider if the local/global distinction becomes central to your argument.]
- **Dasgupta, Liu, Fialkov, et al. (2026)**, arXiv:2604.11542 — stochastic Pop III XRB luminosities in 21cmSPACE. Finds “The impact of stochasticity on the global 21-cm signal and on the large-scale power spectrum is found to be negligible.” Orthogonal to Douna et al.

Assessed from ADS abstracts, found supporting, not downloaded

Two further papers surfaced by the ADS metallicity search were checked and found to *corroborate* H1, so no PDF was retrieved:

- **Fornasini, Kriek, Sanders, et al. (2019)**, ApJ **885**, 65 (MOSDEF, $z \sim 2$): “Splitting our sample by Z , we find that L_X /SFR and Z are anticorrelated with 97% confidence.” They add that “the statistical significance of the observed trends is weak owing to the low X-ray statistics.”
- **Fornasini, Civano & Suh (2020)**, MNRAS **495**, 771: “We find no significant variation of the L_X –SFR– Z relation with redshift. Our results provide further evidence that the Z dependence of HMXBs is responsible for the redshift evolution of L_X /SFR.”

[Assessed at abstract level only.] If H1 becomes load-bearing in your paper, read both in full — see the Sazonov & Khabibullin (2017) reversal above for why.

6 Global assessment and confidence flags

6.1 Summary table

Paper	Target	Strength	Claim(s) contradicted
Dhandha et al. (2025)	HYPOTHESIS	direct	H4 ($T_K(z=10) \lesssim 66$ K); R3
HERA Collab. (2023)	HYPOTHESIS	direct	H1 (warm IGM); H2 (opposite sense); R3
Leite et al. (2017)	RESULTS	direct	R3, R4 (ionization is negligible)
Gessey-Jones et al. (2023)	RESULTS	direct	R4, R5, R7 (global averaging invalid)
Owen et al. (2018)	HYPOTHESIS	direct	H6 (no B field); R1, R2, R7
Jana & Nath (2018)	RESULTS	moderate	R3 (patchy); H6, H8
Yokoyama & Ohira (2023)	HYPOTHESIS	moderate	§2.1 process list; H6, H7
Owen et al. (2019)	RESULTS	moderate	R4, R5, R7 (deposition range)
Sartorio et al. (2023)	HYPOTHESIS	moderate	H5 (L_k uncertainty); H1
Lehmer et al. (2021)	HYPOTHESIS	moderate	H1 (only at $L > 10^{38}$)
Kaur et al. (2022)	HYPOTHESIS	moderate	H1 (factor ~ 3 on T_{IGM})
Sotomayor Checa et al. (2021)	HYPOTHESIS	moderate	H3, H5 (jet may not emerge)
Yang et al. (2024)	RESULTS	indirect	H3 (CRe energetically minor)

Paper	Target	Strength	Claim(s) contradicted
Bera et al. (2023)	HYPOTHESIS	direct	H2, H3; missing Alfvén channel
Singh et al. (2022)	HYPOTHESIS	direct	EDGES context only

6.2 What the audit actually establishes

1. **No published rebuttal of Douna et al. (2018) exists.** All 18 papers citing it — the ADS list, which is a strict superset of what Semantic Scholar, OpenAlex, OpenCitations and INSPIRE-HEP return — cite it neutrally or approvingly. No author has challenged its Monte Carlo, its escape fractions, or its rate estimates in print. [confidence: high] — this rests on the authoritative citation list and a full reading of the citation context in each.
2. **The strongest conflict is with a premise, not a calculation.** Dhandha et al. (2025) infer $T_K(z = 10) \lesssim 66$ K at 95% credibility, and HERA (2023) measures $T_K < 313.2$ K at $z = 10.4$ (95%), against the assumed $T_{\text{IGM}} \sim 10^4$ K. Douna et al. bracketed this themselves and stated the direction of the effect, so the paper is not invalidated — but the headline number R3 ($f_{\text{ion}} \sim 0.1$) is quoted from the warm branch, and the observationally permitted branch is the cold one. [confidence: high on the temperature comparison; medium on the propagated effect on f_{ion} , because I did not recompute $\alpha(T)$ or re-run the balance; and Dhandha et al. explicitly exclude flexible Pop III source models, which is the regime Douna et al. occupy.] **This is the constraint your paper most needs to address head-on.**
3. **The second-strongest conflict is with the transport model.** Owen et al. (2018) on magnetic confinement, Jana & Nath (2018) and Gessey-Jones et al. (2023) on the consequences of diffusion, and Yokoyama & Ohira (2023) and Bera et al. (2023) on channels that only exist in a magnetized medium, jointly indicate that H6 — deferring magnetic fields to a companion paper — is the load-bearing simplification. [confidence: medium-high] — the papers agree that B matters and *disagree with each other* on whether the result is patchy or uniform heating.
4. **The ionization claim is the most exposed result.** Leite et al. (2017) obtain a rate of the same order as Douna et al. and call it negligible. Nothing published since has revisited this. [confidence: high that the disagreement in interpretation is real; low that it is resolvable from the two papers alone.]
5. **Carrier mismatch is pervasive and is the main weakness of this audit.** Ten of the thirteen entries model CR *protons* or X-ray photons. Douna et al. model CR *electrons*. Arguments that transfer cleanly are the geometric/transport ones (rigidity-based confinement, deposition range) and the observational ones (IGM temperature). Arguments about energy budget and spectrum do not transfer without a calculation neither paper performs. [This is stated as a limitation of the literature, not of the search: no paper modelling CR-electron transport from high- z microquasars other than Douna et al. (2018) and Tueros et al. (2014) was found.]

6.3 Explicitly not verified

- No numerical quantity from any paper was independently recomputed. Every number quoted is as printed by its authors. No cross-check with `sympy` or by limiting case was performed, because no derivation was reproduced. [no cross-check performed]
- The comparison of HERA’s T_K bound at $z = 10.4$ with Douna et al.’s assumption at $z = 10$ ignores the (small, monotonic) redshift difference; the direction of the inequality is unaffected but the factor is approximate.
- Search recall *was* validated against NASA ADS (Section 1.3), which added four citing papers and two conflicting ones. ADS full-text search (`full:`) covers only the fraction of the corpus

for which ADS holds full text; a conflicting statement buried in the body of a paper outside that fraction would still be missed.

- Two papers (Fornasini et al. 2019, 2020) were assessed from their ADS abstracts only and judged supporting. The Sazonov & Khabibullin (2017) reversal in Section 5 shows abstract-level screening can invert on full reading, so this is a real residual risk.
- Excerpts were extracted mechanically and hand-checked against the printed form for mathematical symbols only. Long quotations were not re-read against the published (journal) versions; all are from the arXiv PDFs stored in `against/`.

Running reference list

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